

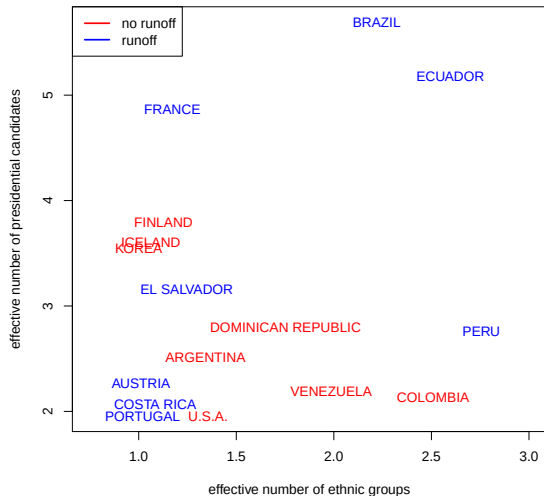
Lecture 15

October 31, 2023

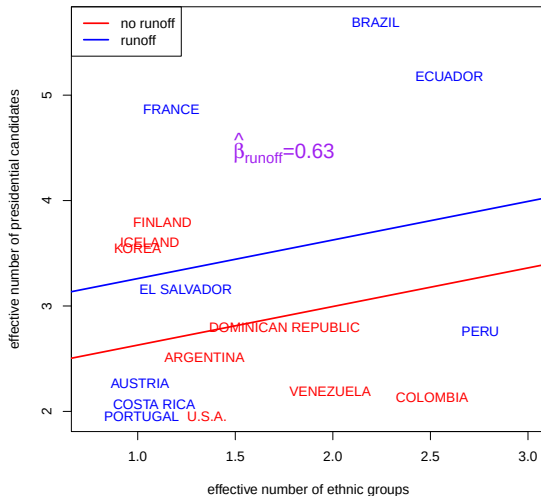
Multiple Regression with Interactions and Polynomials

- ▶ For the last few lectures we have focused on linear regression coefficients.
- ▶ Linear (with respect to the betas) regression can be made non-linear (with respect to the covariates) with the inclusion of polynomial terms.
- ▶ Even with a binary covariate of interest, polynomials for the controls and interactions make the goal of linear regression unclear.
- ▶ Average Partial Derivatives are an important approach (analogous to the “average over people” estimator from earlier in the term).

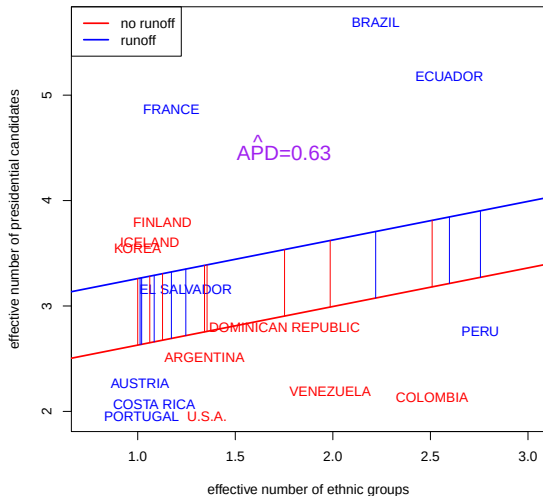
Neto and Cox (1997) Table 3 Data



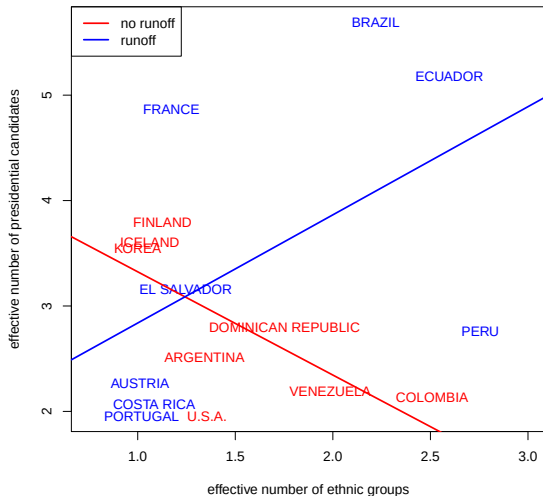
Additive Linear Regression



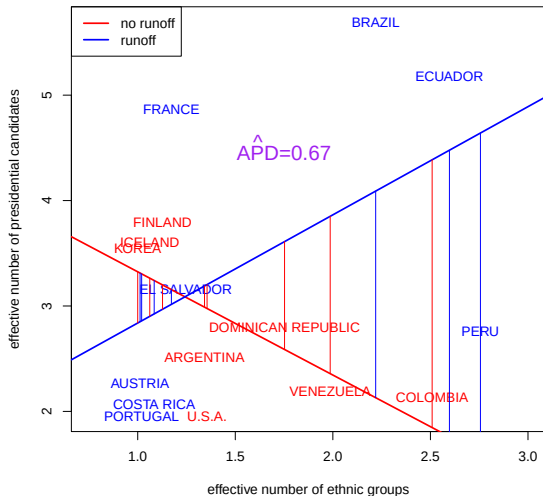
Average Partial Derivative (Runoff) in Additive Model



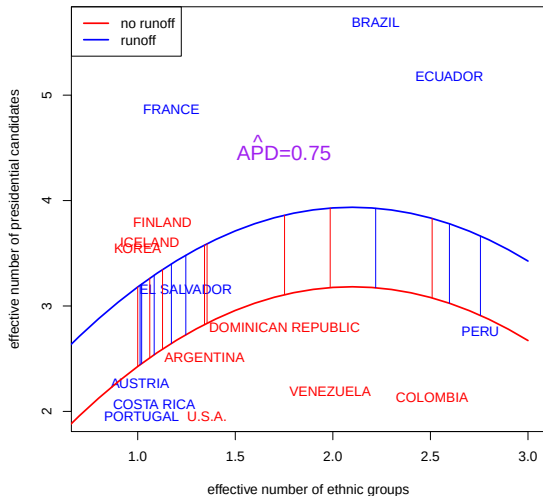
APD in Interactive/Unpooled Linear Model



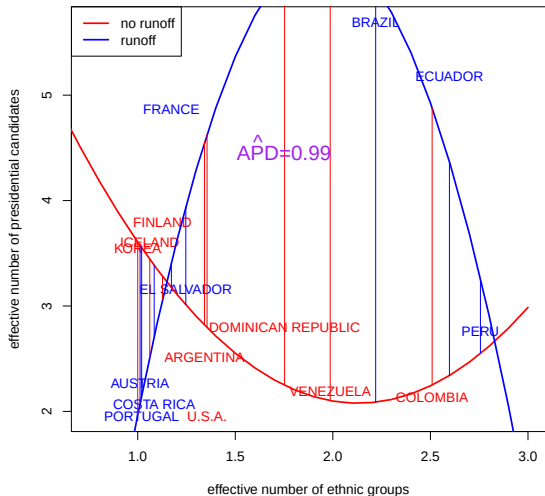
APD in Interactive/Unpooled Linear Model



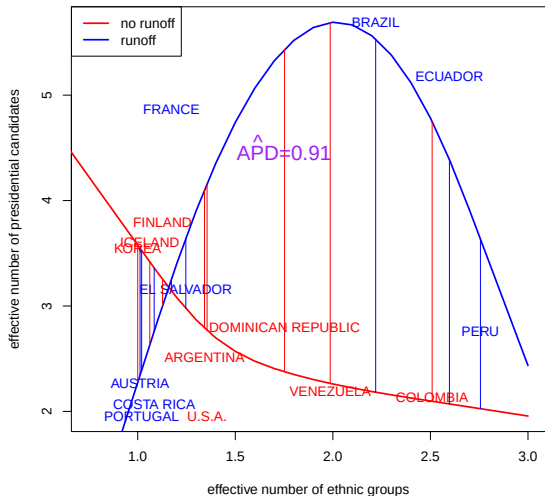
APD in Additive Quadratic Models



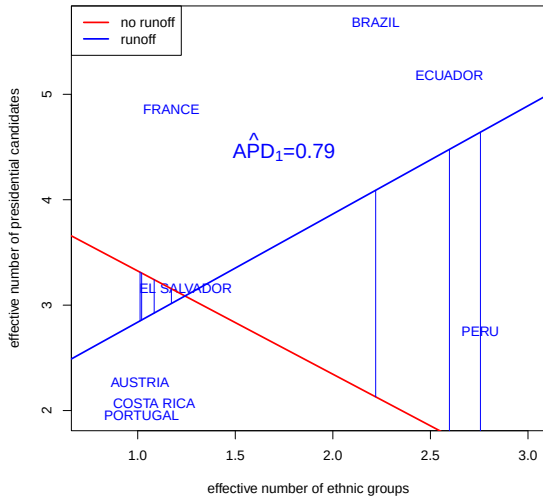
APD in Interactive/Unpooled Quadratic



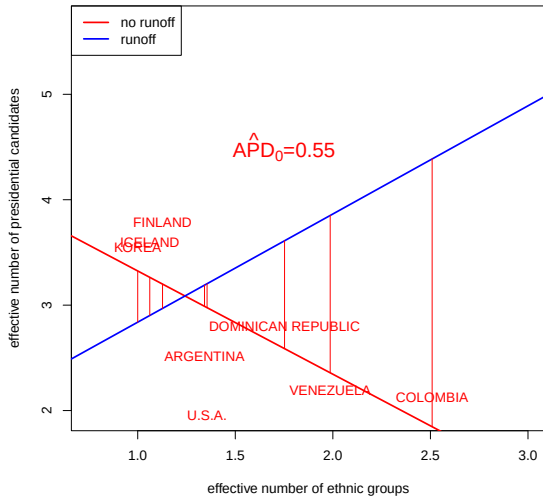
APD in Unpooled Local Regression



APD (Runoff) for the Runoff Countries in Interactive Linear Model



APD (Runoff) for the Non-Runoff Countries in Interactive Linear Model



Mathematical Presentation

What if the CEF is nonlinear?

I.e., what if $E[Y|X] \neq a + bX$?

Can we still use linear regression?

When the CEF is linear in X , X^2 ,

$$E[Y|X] = \beta_0 + \beta_1 X + \beta_2 X^2.$$

How to interpret?

Before: linear estimate of slope of CEF was β_1 .

How can we generalize?

When the CEF is linear in X , X^2 ,

$$\frac{\partial E[Y|X]}{\partial X} = \frac{\partial(\beta_0 + \beta_1 X + \beta_2 X^2)}{\partial X}$$

When the CEF is linear in X , X^2 ,

$$\frac{\partial E[Y|X]}{\partial X} = \beta_1 + 2\beta_2 X$$

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$$\frac{\partial E[Y|X]}{\partial X} = \beta_1 + 2\beta_2 X$$

The slope of the CEF depends on the value of X .

What about multiple X :

- ▶ Additive, Interactive, and Saturated Models
- ▶ with polynomials
- ▶ What if not linear in β ?

What happens when the conditional slope of the CEF with respect to X_1 depends on the value of X_2 ?

The BLP is still the BLP, but we might be missing out on an interesting *interaction* between the variables X_1 and X_2 .

So, to formalize a bit, what if the CEF is otherwise linear, but includes an *interaction*?

$$E[Y|X_1, X_2] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

Much as before, we could derive the slope of the CEF with respect to a given variable.

$$\frac{\partial E[Y|X_1, X_2]}{\partial X_1} = \frac{\partial \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2}{\partial X_1}.$$

Much as before, we could derive the slope of the CEF with respect to a given variable.

$$\frac{\partial E[Y|X_1, X_2]}{\partial X_1} = \beta_1 + \beta_3 X_2.$$

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The conditional slope of the CEF with respect to X_1 *depends on the value of* X_2 .

For full generality, interactions can be combined with polynomials, e.g.:

$$E[Y|X_1, X_2] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \beta_4 X_2^2 + \beta_5 X_1^2 X_2$$

With slopes computed by taking partial derivatives.

Let's find

$$\frac{\partial E[Y|\mathbf{X}]}{\partial X_1} \Big|_{\mathbf{x}=\mathbf{x}}$$

Average Partial Derivative

The partial derivative can differ for all possible $\mathbf{X} = x$

$$\frac{\partial E[Y|\mathbf{X}]}{\partial X_1} \Big|_{\mathbf{x}=x}$$

Often, we will be interested in an average of these partials

$$APD_{X_1} = E \left[\frac{\partial E[Y|\mathbf{X}]}{\partial X_1} \Big|_{\mathbf{x}=x} \right]$$

where the average is taken over the distribution of $\mathbf{X}$.

Estimated Average Partial Derivative

In practice, we just average over the partials at the data points

$$\widehat{APD}_{X_1} = \frac{1}{n} \sum_{i=1}^n \left[\frac{\partial E[Y|\mathbf{X}]}{\partial X_1} \Big|_{\mathbf{x}=\mathbf{x}_i} \right]$$

Estimated APD Examples on board:

Single X examples

- ▶ Linear
- ▶ Quadratic
- ▶ Cubic

X_1, X_2 examples

- ▶ Linear Additive
- ▶ Linear Interactive
- ▶ Polynomial with Interactions

Recall the estimated APD:

$$\widehat{APD}_{X_1} = \frac{1}{n} \sum_{i=1}^n \left[\frac{\partial \hat{E}[Y|\mathbf{X}]}{\partial X_1} \Big|_{\mathbf{x}=\mathbf{x}} \right]$$

What if we had a binary regressor, X_1 (i.e., $\text{Supp}(X_1) = \{0, 1\}$)?

Recall the estimated APD:

$$\widehat{APD}_{X_1} = \frac{1}{n} \sum_{i=1}^n \left[\frac{\partial \hat{E}[Y|\mathbf{X}]}{\partial X_1} \Big|_{\mathbf{x}=\mathbf{x}_i} \right]$$

What if we had a binary regressor, X_1 (i.e., $\text{Supp}(X_1) = \{0, 1\}$)?

$$\widehat{APD}_{X_1} = \frac{1}{n} \sum_{i=1}^n [\hat{E}[Y|X_1 = 1, \mathbf{X}_{(-1)} = \mathbf{x}_{i,(-1)}] - \hat{E}[Y|X_1 = 0, \mathbf{X}_{(-1)} = \mathbf{x}_{i,(-1)}]]$$

Note that this is exactly what we did with the Neto and Cox (1997) example!

Confidence Intervals for APDs

Just Bootstrap!